



SCHOOL OF COMPUTER SCIENCE ENGINEERING

ASSIGNMENT No. 1

SEMESTER & CLASS: 5th & A/B

FACULTY NAME: Dr. Puneet Bawa

COURSE NAME: Introduction to Machine Learning

COURSE CODE: CSE2505

START DATE: 07-09-2026

DURATION (IN CASE OF TEST/QUIZ ETC.)

DATE OF CONDUCTION (IN CASE OF TEST/QUIZ ETC.) : NA

SUBMISSION DATE: 21-09-2026

Max. Marks: 20

Ques. No	Question	CO	Marks																														
Q1.	a) Differentiate between Artificial Intelligence, Data Science and Machine Learning. b) Identify whether each application primarily uses supervised, unsupervised or reinforcement learning: - Email spam detection - Customer segmentation - Robot navigation - House-price prediction	CO1	4																														
Q2	A university wants to develop a model to predict whether a student is at risk of failing a course. a) Design a complete machine learning workflow for this problem, from problem definition to model evaluation. b) Suggest four relevant data attributes and two suitable data-collection techniques. Mention one ethical concern associated with collecting student data.	CO1	4																														
Q3.	Consider the following dataset: <table><tr><th>Student</th><th>Study Hours</th><th>Attendance (%)</th><th>Assignment Score</th><th>Final Result</th></tr><tr><td>A</td><td>4</td><td>85</td><td>78</td><td>Pass</td></tr><tr><td>B</td><td>6</td><td>Missing</td><td>82</td><td>Pass</td></tr><tr><td>C</td><td>5</td><td>88</td><td>80</td><td>Pass</td></tr><tr><td>D</td><td>40</td><td>42</td><td>35</td><td>Fail</td></tr><tr><td>E</td><td>3</td><td>75</td><td>Missing</td><td>Pass</td></tr></table>	Student	Study Hours	Attendance (%)	Assignment Score	Final Result	A	4	85	78	Pass	B	6	Missing	82	Pass	C	5	88	80	Pass	D	40	42	35	Fail	E	3	75	Missing	Pass	CO1	4
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	a) Identify the missing values and possible outlier. Justify your answer. b) Recommend appropriate methods for handling the missing values and outlier. c) Explain whether normalization or standardization should be applied before model training.		
Q4	For a house-price prediction model, the available features are: <i>Area, Number of Bedrooms, Construction Year, Location, Owner Name, and Selling Price</i> a) Identify the target variable and distinguish numerical and categorical features. b) Create one meaningful engineered feature using the available data. c) Recommend any two EDA techniques for studying the relationship between the features and selling price. d) Identify one feature that may be removed and justify your decision.	CO2	4
Q5	A simple linear regression model is: $\hat{y} = 25 + 8x$ where x represents study hours and $\hat{y}$ represents the predicted examination score. a) Predict the score of a student who studies for 5 hours. b) Interpret the slope and intercept of the model. c) Differentiate between simple linear regression, multiple linear regression and logistic regression based on their inputs and outputs. d) Name two suitable evaluation metrics for a regression model and explain the purpose of train-test split or cross-validation.	CO2	4

COURSE OUTCOME:

CO1	Understand machine learning basics to articulate or demonstrate and pinpoint applicable problem scenarios for effective student problem-solving	K2
CO2	Understand regression analysis fundamentals to suggest predictive models from data and evaluate their effectiveness in value prediction.	K2
CO3	Demonstrate unsupervised machine learning strategies to address clustering challenges and assess their efficacy in performance analysis.	K4
CO4	Evaluate Illustrate the principles of Mining Frequent Patterns, Associations, and Data Streams, applying them to solve pertinent issues and evaluate performance. Identify applicable problems for student application, analysing effectiveness and limitations, emphasizing lifelong learning significance. .	K5